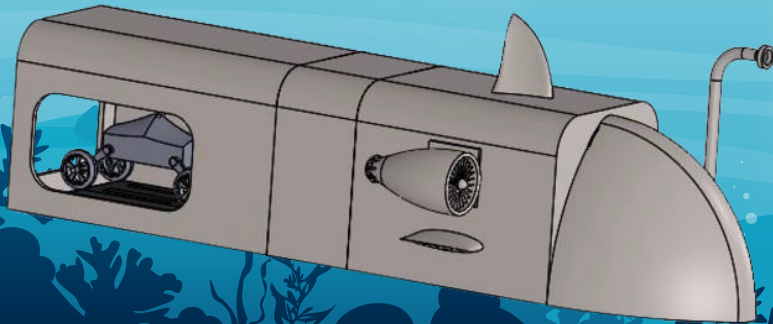


Sea Splicer

ME 1670 A Team 10 – CTRL + SEA

Ian Ayala
Carter Canfield
Alexander Dorsey
Aneesh Kumar
Sahaj Patel



Outline

- 1) Motivation
- 2) Individual Part Models
- 3) Assembly Models
- 4) Challenges
- 5) Rendered Views
- 6) Animation



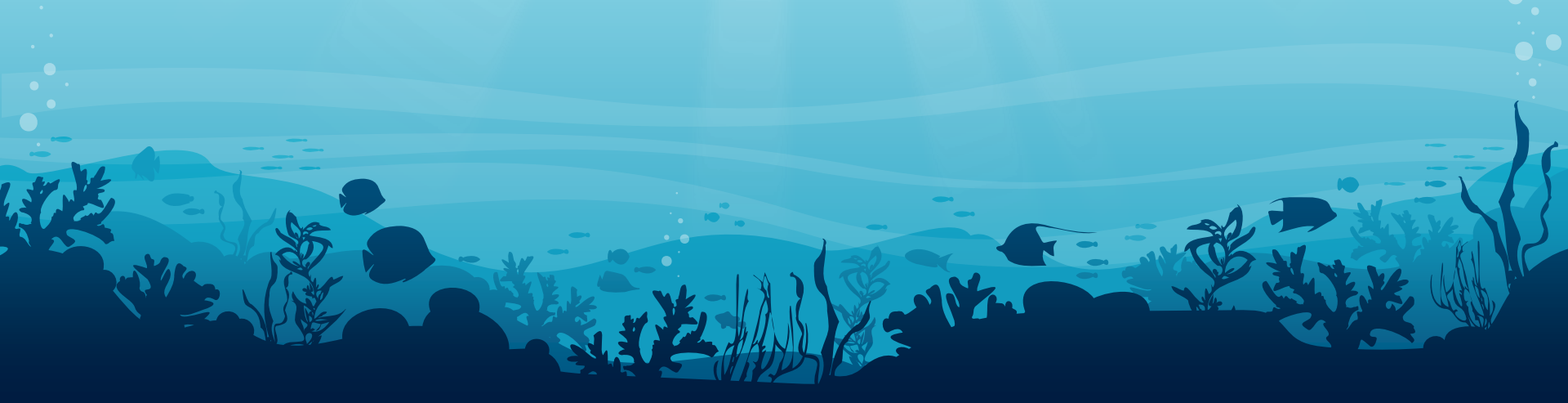
Motivation

Motivation/Purpose:

- Futuristic maritime exploration, research, and living
- Designed around the size of a 6-foot person

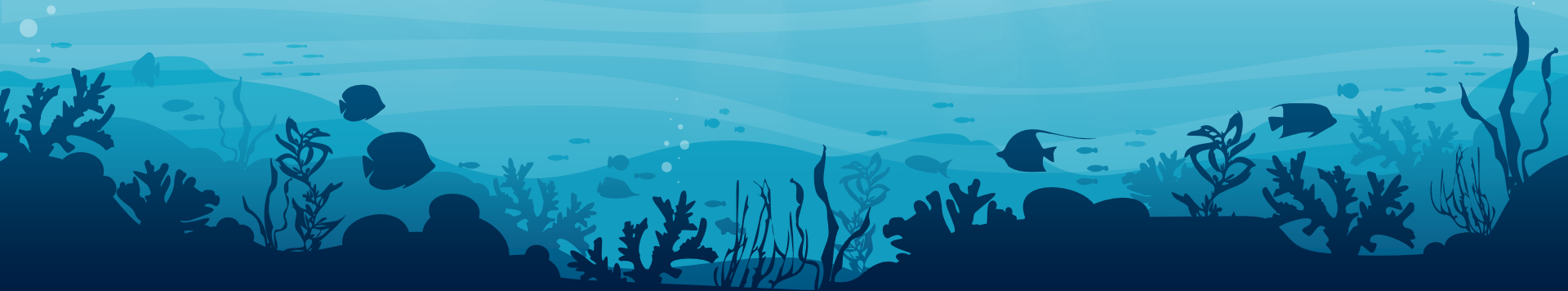
External Reference:

- Subnautica Sea Truck

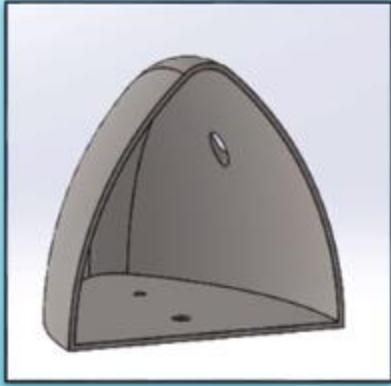


1. Cockpit

Sahaj Patel



Individual Parts



Cockpit Shell



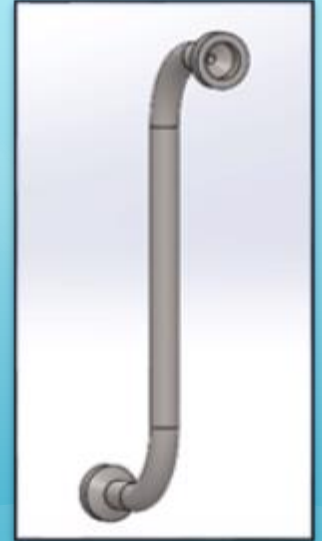
Cockpit Chair



Cockpit Door

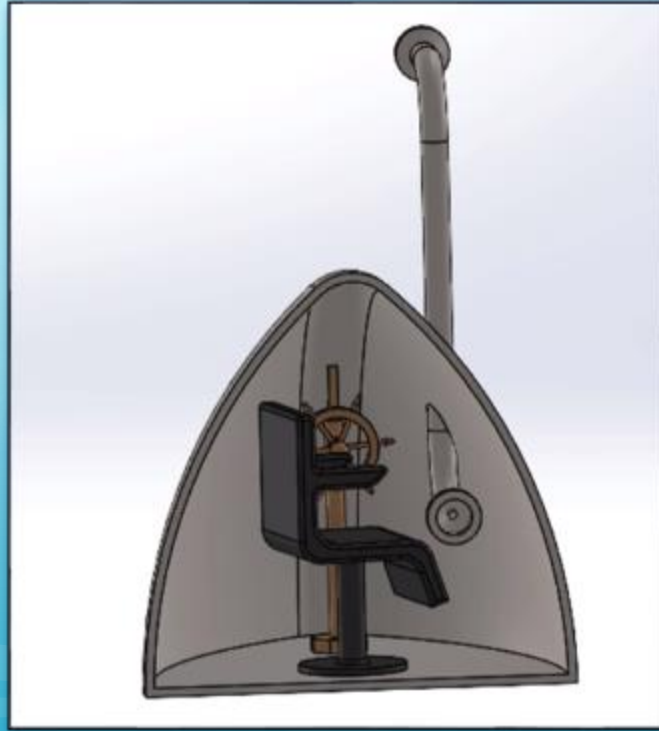


Cockpit Steering
Wheel Mount



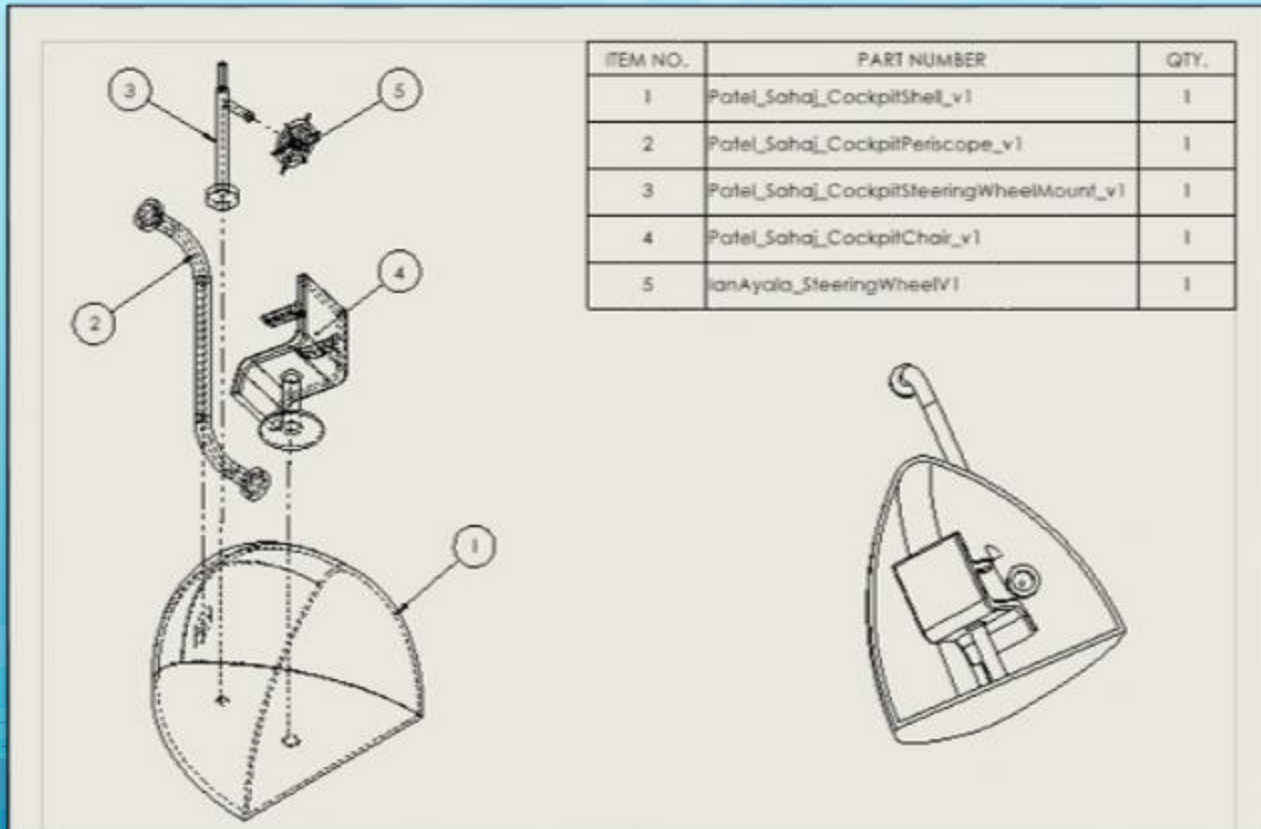
Cockpit Periscope

Individual Sub-Assemblies



Cockpit Sub-Assembly

Individual Exploded Views



The background is a stylized underwater scene. At the top, several bright sun rays beam down from the surface. The water is depicted with horizontal wavy bands of different shades of blue. In the foreground, there is a dark blue silhouette of the ocean floor, featuring various coral reefs, seaweed, and small rocks. Several small, dark blue fish are scattered throughout the scene, swimming in different directions. Bubbles are shown as small white circles rising from the bottom and floating in the water.

2. Main Compartment

Ian Ayala

Individual Parts



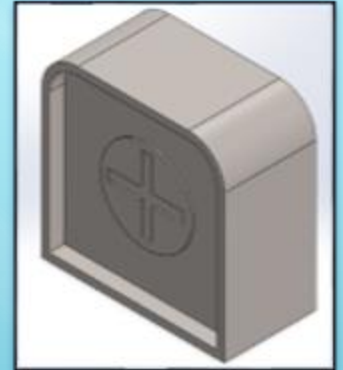
Cockpit Steering
Wheel



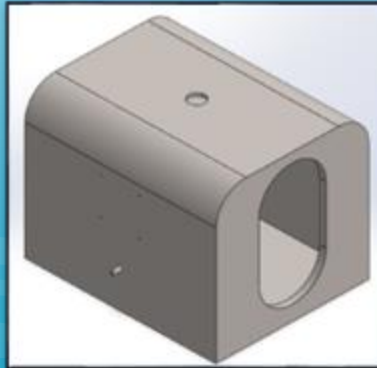
Bed



Car Wheel

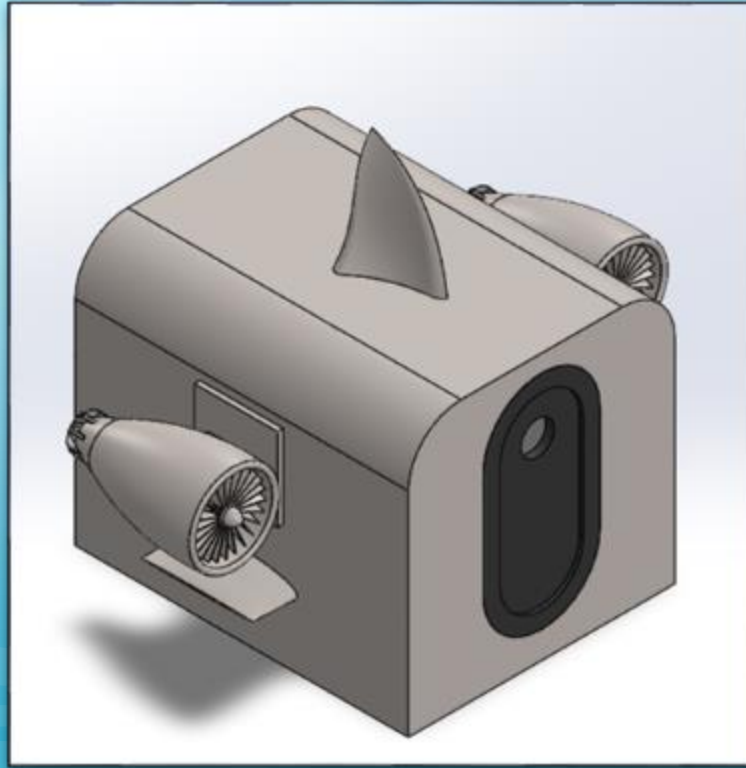


Airlock



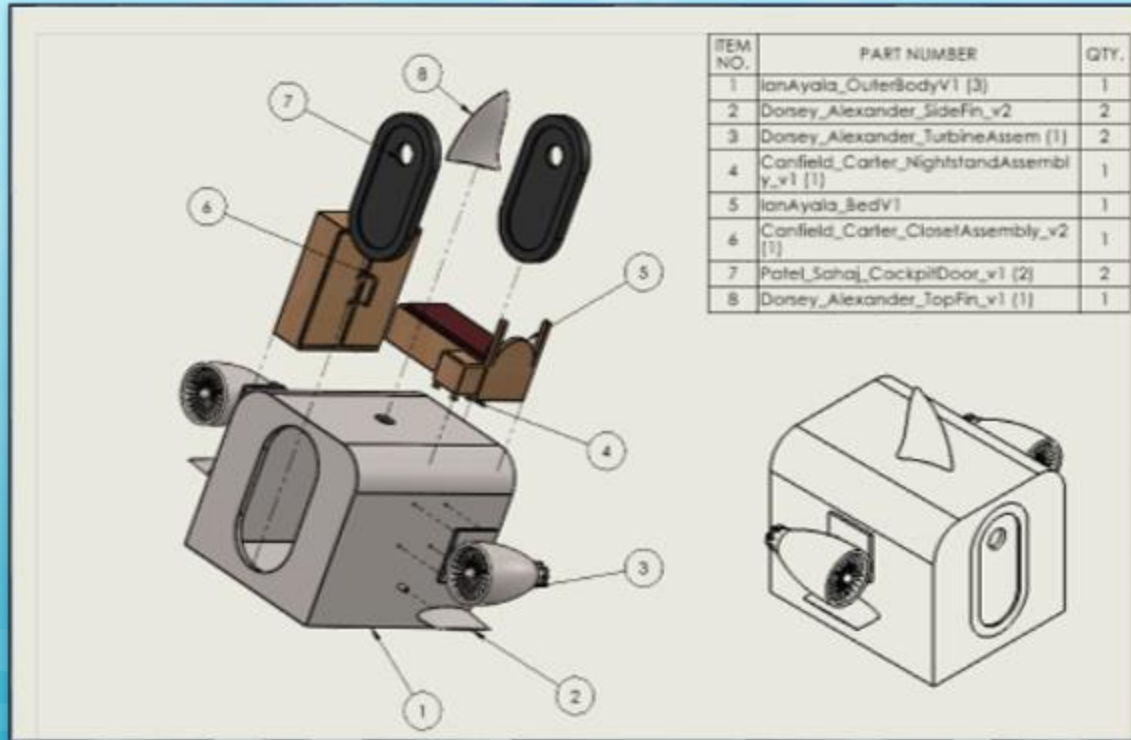
Compartment Body

Individual Sub-Assemblies



Main Compartment Sub-Assembly

Individual Exploded Views

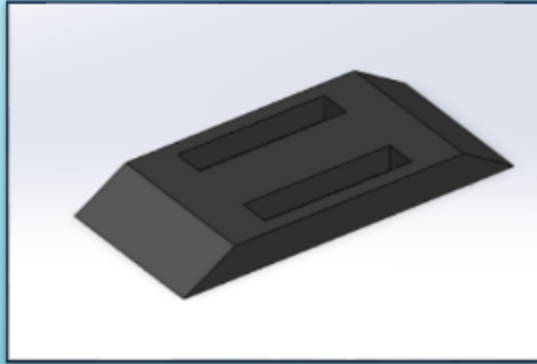


The background is a light blue gradient with several white sun rays emanating from the top center. The bottom of the image features a dark blue silhouette of an underwater landscape with various coral reefs, seaweed, and small fish. Bubbles are scattered throughout the scene, particularly on the left and right sides.

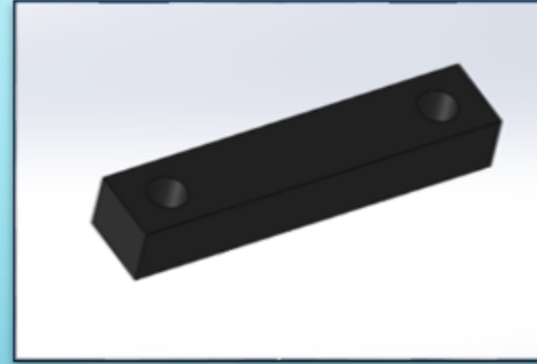
3. Bedroom

Carter Canfield

Individual Parts



Car Platform Base



Car Platform Lift



Middle Tube



Inner Tube



Outer Tube

Individual Parts (continued)



Left Door



Closet Body



Right Door

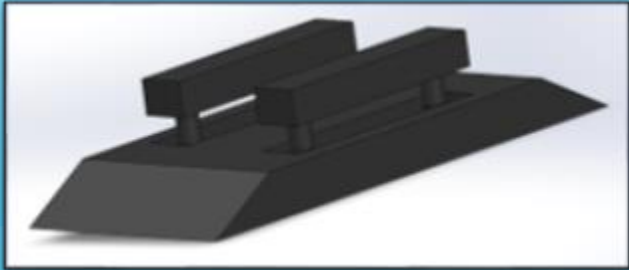
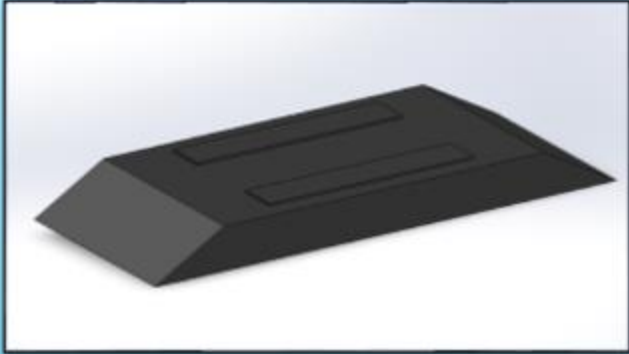


Nightstand Body



Nightstand Drawer

Individual Sub-Assemblies

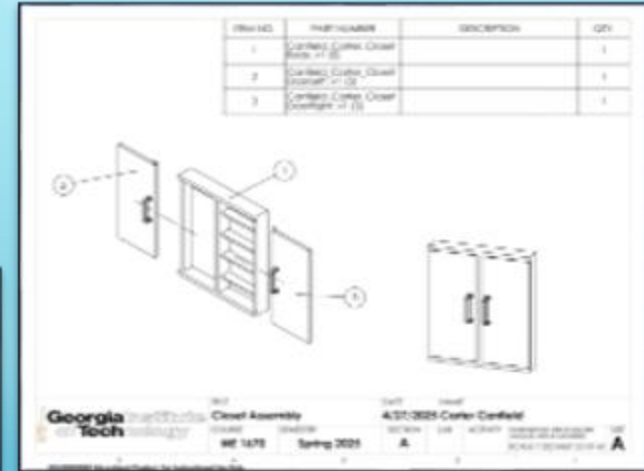
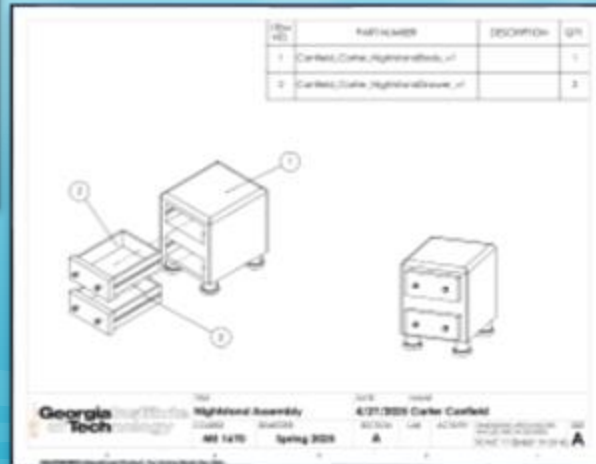
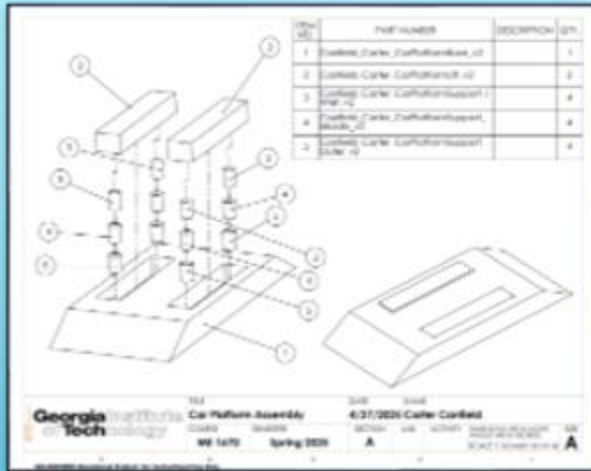


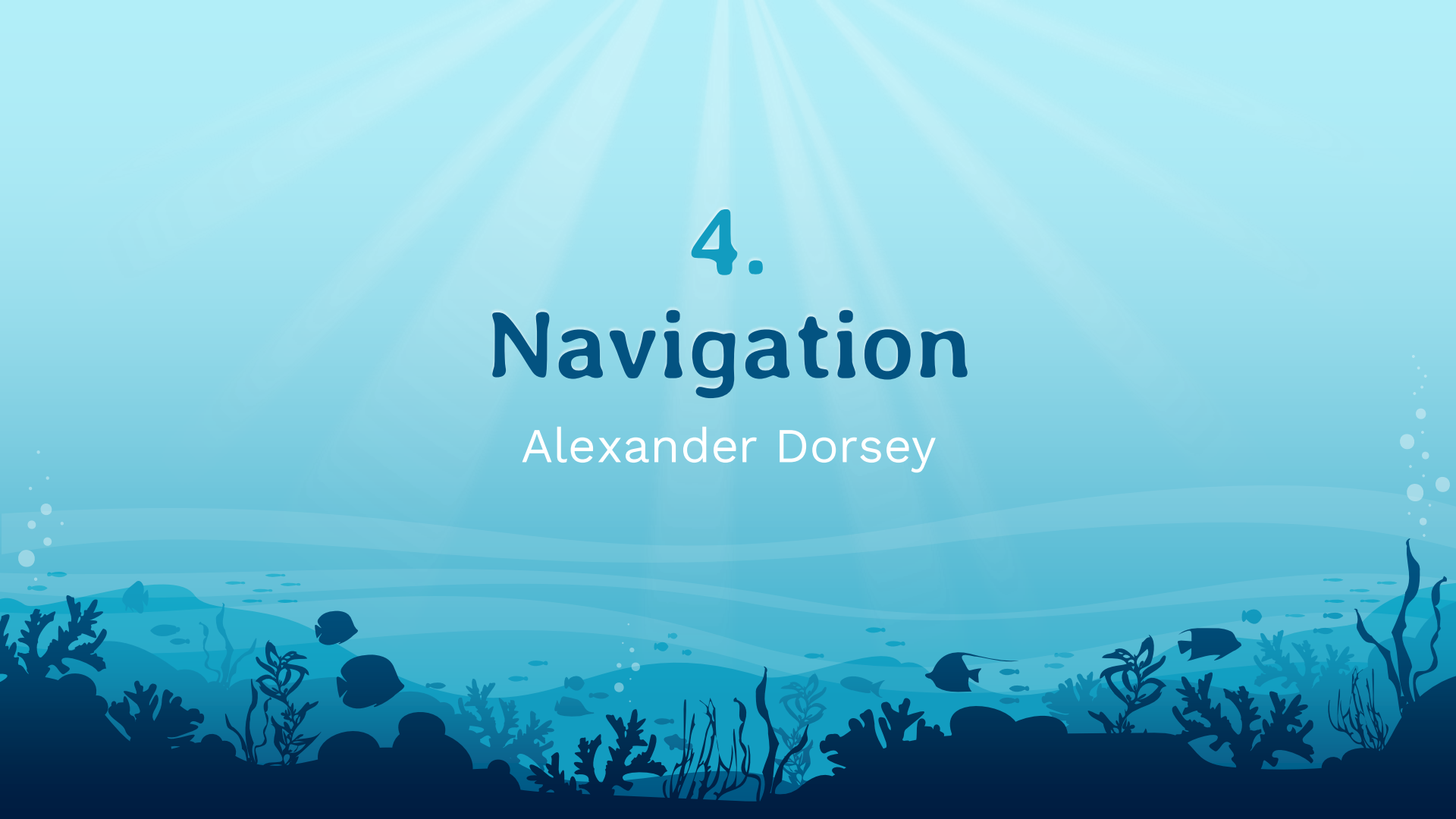
Car Platform Sub-Assembly

Closet Sub-Assembly

Nightstand Sub-Assembly

Individual Exploded Views

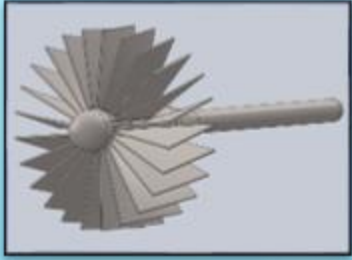


The background is a stylized underwater scene. At the top, several bright blue sun rays fan out from the center towards the edges. The water is represented by horizontal bands of varying shades of blue. At the bottom, there is a dark blue silhouette of the ocean floor with various coral reefs and sea anemones. Several small, dark blue fish are scattered throughout the scene, swimming in different directions. Bubbles are depicted as small white circles, rising from the bottom and floating in the water.

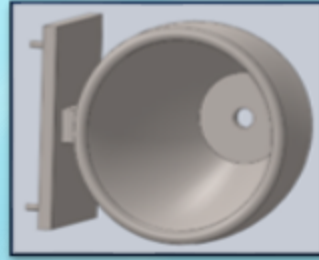
4. Navigation

Alexander Dorsey

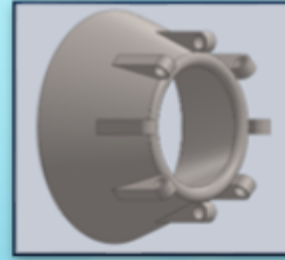
Individual Parts



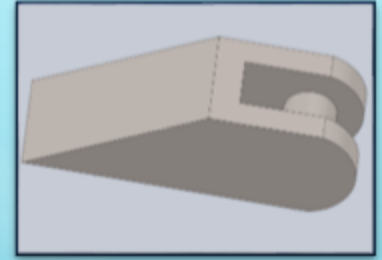
Turbine Blades



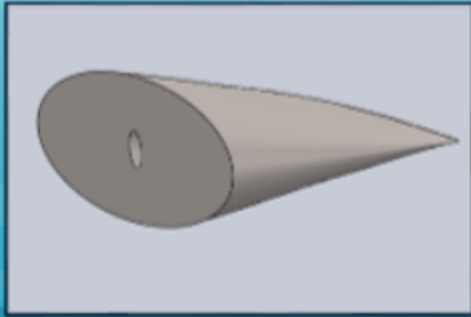
Turbine Main Cover



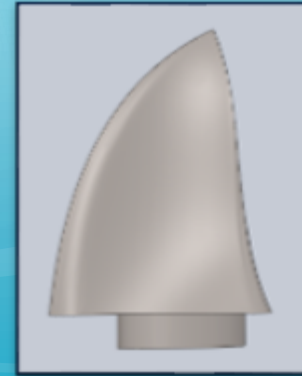
Turbine Output Nozzle



Turbine Nozzle Flaps

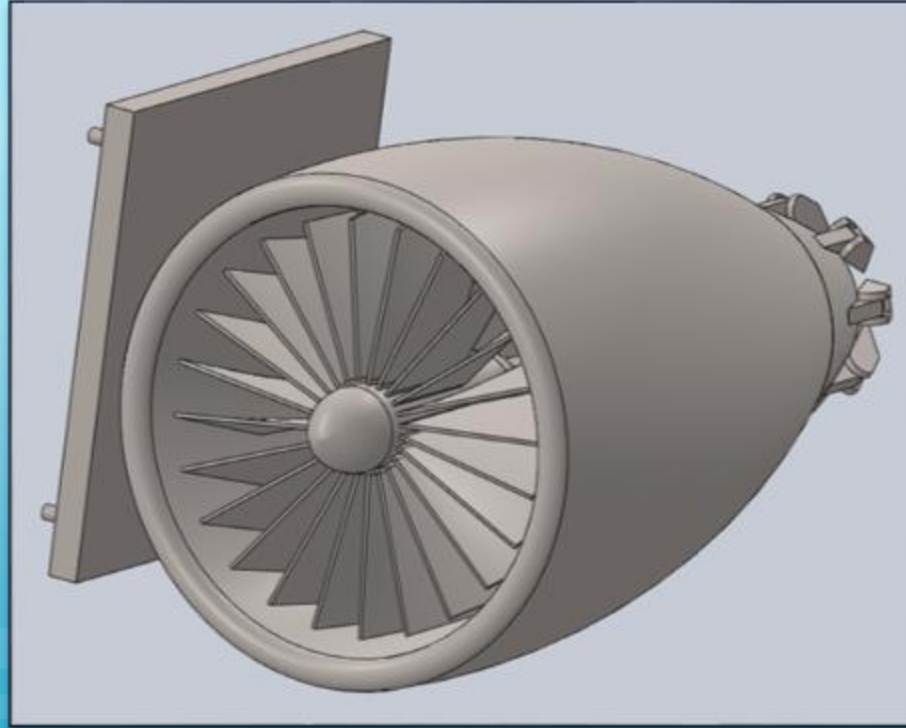


Side Fin



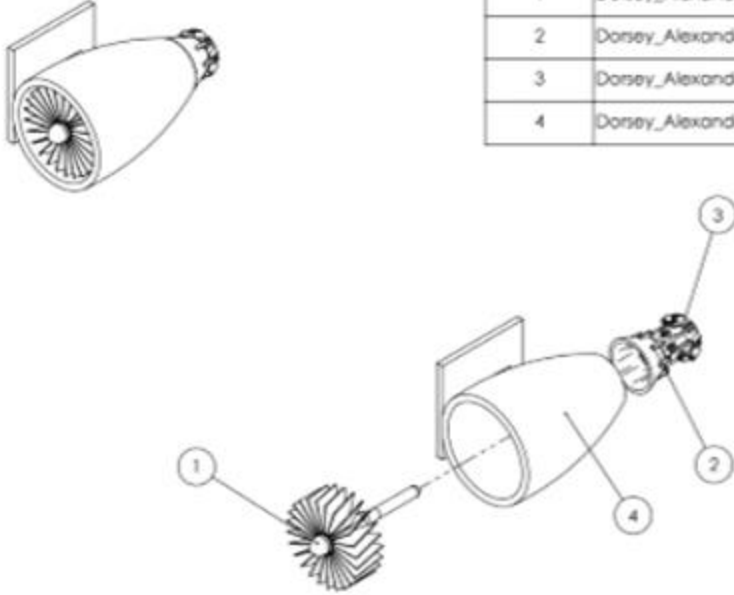
Top Fin

Individual Sub-Assemblies



Turbine Sub-Assembly

Individual Exploded Views



ITEM NO.	PART NUMBER	QTY.
1	Dorsey_Alexander_Blades_v1	1
2	Dorsey_Alexander_OutputNozzle_v1	1
3	Dorsey_Alexander_NozzleFlap_v1	8
4	Dorsey_Alexander_MainCover_v2	1



TITLE Turbine Assembly (Exploded)		DATE 4/27/2025	NAME Alexander Dorsey	
COURSE ME 1670	SEMESTER Spring 2025	SECTION A	LAB	ACTIVITY

DIMENSIONS ARE IN INCHES
ANGLES ARE IN DEGREES
SCALE 3/8"=1" (SEE DET.)

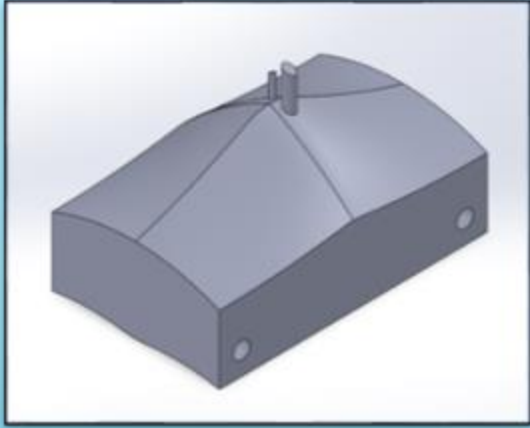
A

The background is a light blue gradient with several white sun rays emanating from the top center. At the bottom, there is a dark blue silhouette of an underwater landscape featuring various coral reefs, seaweed, and several small fish swimming. Bubbles are also visible on the left and right sides of the bottom section.

5. Garage

Aneesh Kumar

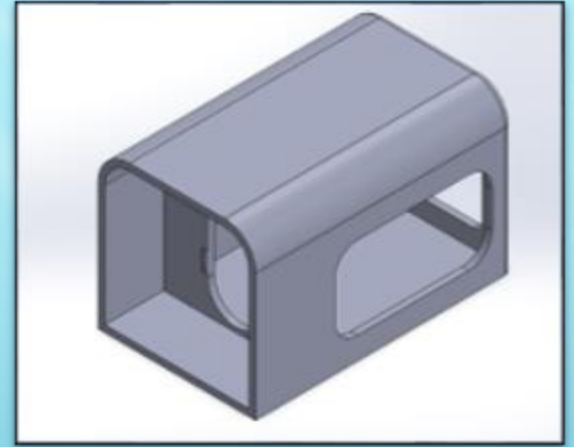
Individual Parts



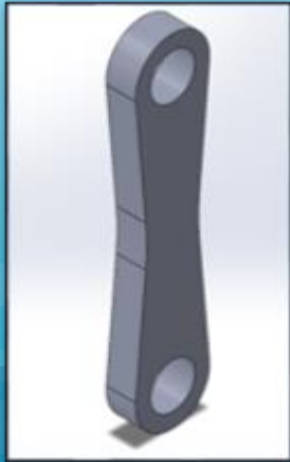
Car Body



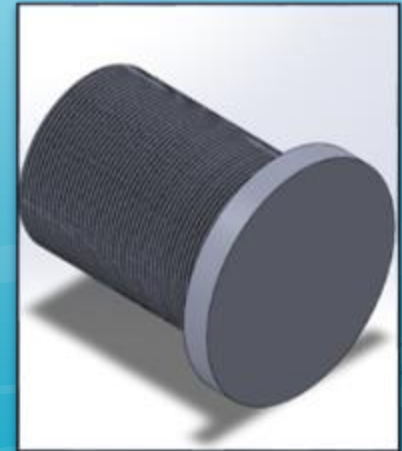
Garage Door



Garage Body

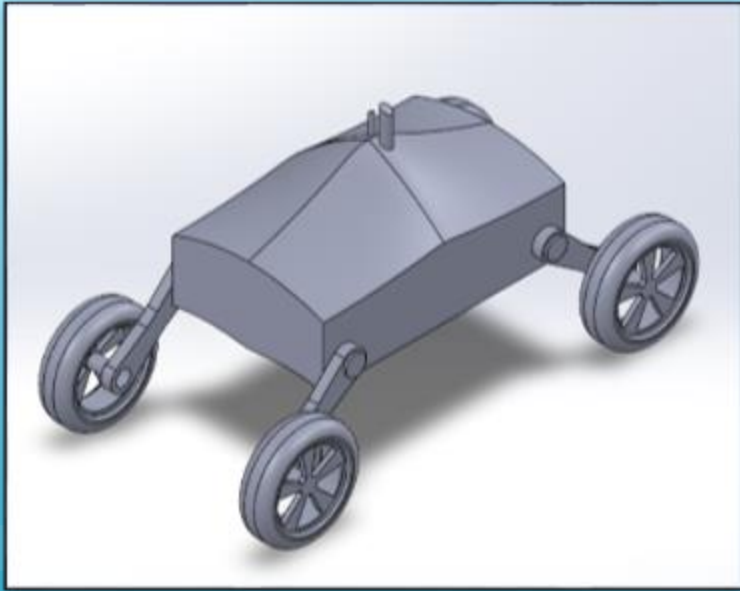


Suspension Rod

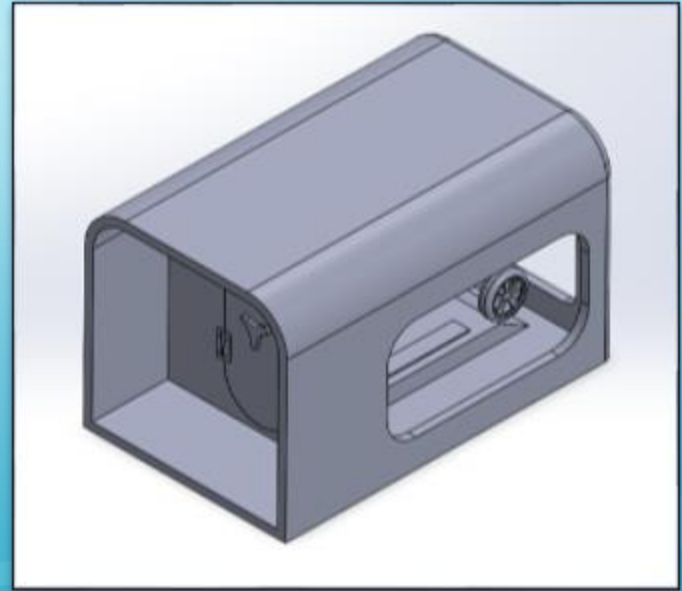


Suspension Bolt

Individual Sub-Assemblies

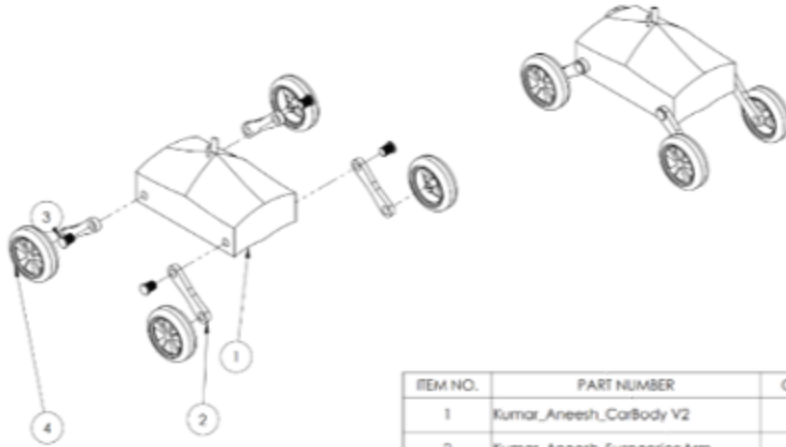


Car Sub-Assembly



Garage Sub-Assembly

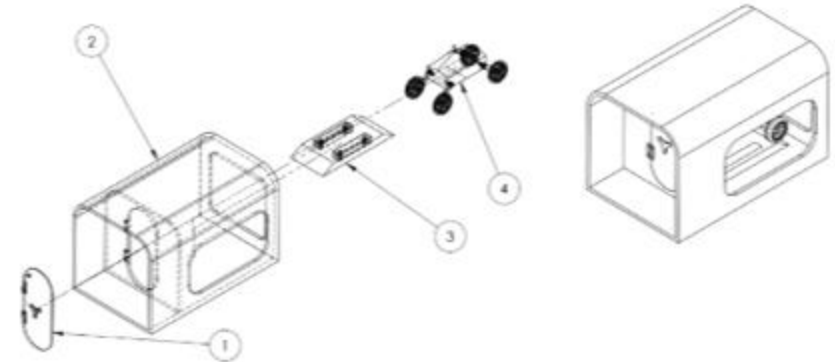
Individual Exploded Views



ITEM NO.	PART NUMBER	QTY.
1	Kumar_Aneesh_CarBody V2	1
2	Kumar_Aneesh_SuspensionArm	4
3	Kumar_Aneesh_SuspensionBolt	4
4	IanAyala_CarWheelV1	4

Georgia Institute of Technology

TITLE: Car Assembly
 COURSE: ME 1670
 SEMESTER: SPRING 2025
 DATE: 4/28/2025
 NAME: Aneesh Kumar
 SECTION: A
 LAB:
 ACTIVITY:
 DIMENSIONS ARE IN INCHES
 ANGLES ARE IN DEGREES
 SCALE: 1:48 SHEET 41 OF 42
 SIZE: A



ITEM NO.	PART NUMBER	QTY.
1	Kumar_Aneesh_GarageDoor	1
2	Kumar_Aneesh_GarageMainBody V2	1
3	Canfield_Carter_CarPlatformAssembly	1
4	Kumar_Aneesh_CarAssemblyV2	1

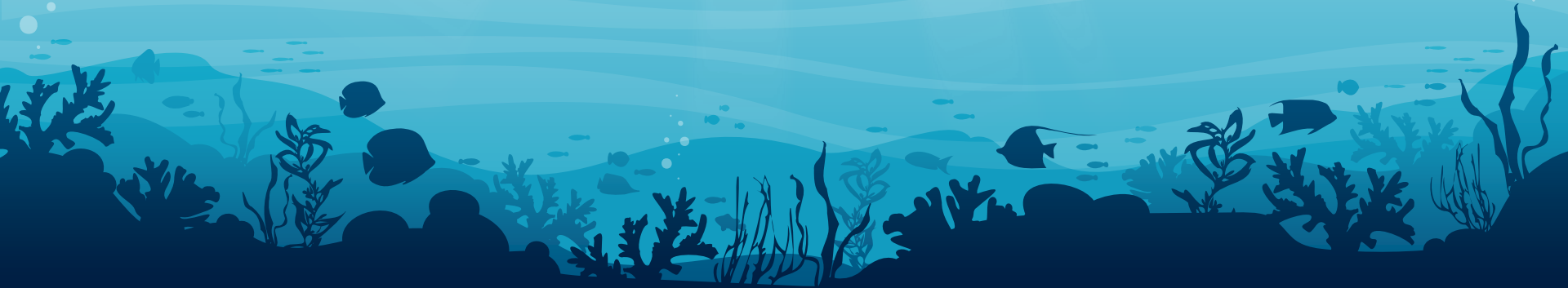
Georgia Institute of Technology

TITLE: Garage Subassembly
 COURSE: ME 1670
 SEMESTER: SPRING 2025
 DATE: 4/27/2025
 NAME: Aneesh Kumar
 SECTION: A
 LAB:
 ACTIVITY:
 DIMENSIONS ARE IN INCHES
 ANGLES ARE IN DEGREES
 SCALE: 1:192 SHEET 30 OF 42
 SIZE: A

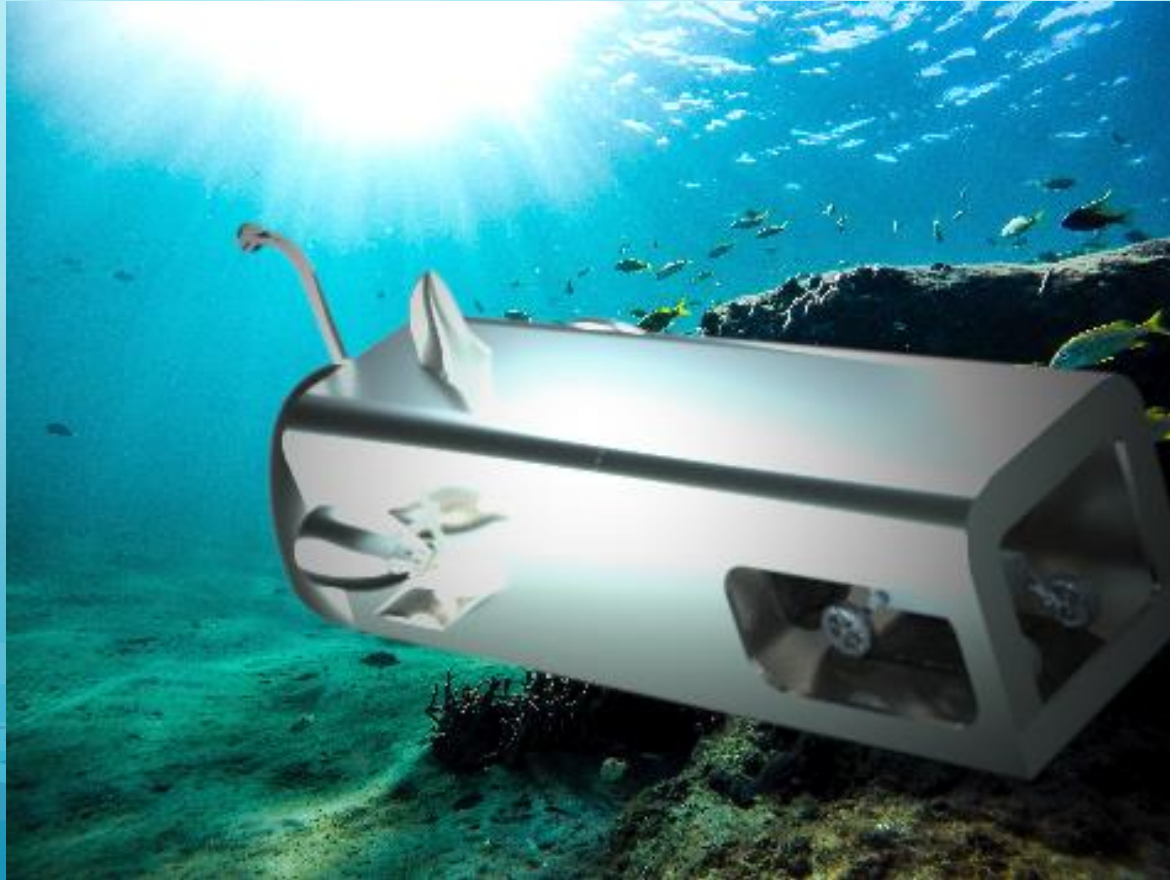
Challenges

- 1) Communication
- 2) Complex Dimensioning
- 3) File Organization
- 4) Time Management

Rendered Views



View 1: Underwater Exploration



View 2: Sea Floor Exploration



View 3: Side Body Detail View



Animation Link (YouTube)

Conclusion

- 1) Overall Experience
- 2) Enjoyable Things
- 3) Frustrating Things
- 4) Lessons Learned

Thank You!

